

Brain Cells

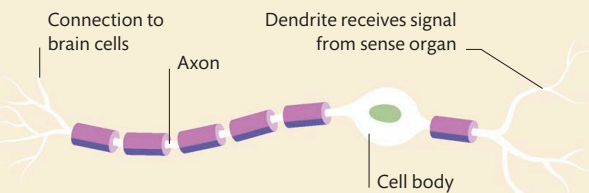
The brain and the rest of the nervous system contains a network of cells called neurons. The role of neurons is to carry nerve signals through the brain and body as electrical pulses.

Neurons

Most neurons have a distinctive branched shape with dozens of filaments, only a few hundred thousandths of a foot thick, extending from the cell body toward nearby cells. Branches called dendrites bring signals into the cell, while a single branch, called the axon, passes the signal to the next neuron. In most cases, there is no physical connection between neurons. Instead, there is a tiny gap, called the synapse, where electrical signals stop. Communication between cells is carried out by the exchange of chemicals, called neurotransmitters (see pp.22–23). However, some neurons are effectively physically connected and do not need a neurotransmitter to exchange signals.

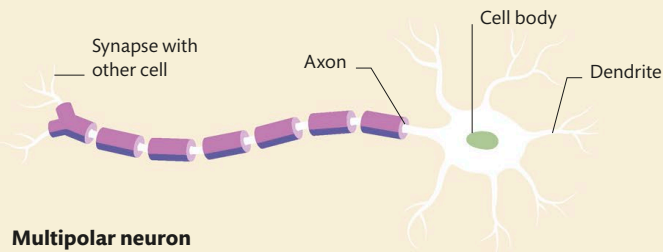
Types of neurons

There are several types of neurons, with different combinations of axons and dendrites. Two common types, bipolar and multipolar neurons, are each suited to particular tasks. Another type of neuron, the unipolar neuron, appears only in embryos.



Bipolar neuron

This type of neuron has one dendrite and one axon. It transmits specialized information from the body's major sense organs.



Multipolar neuron

Most brain cells are multipolar. They have multiple dendrites connecting to hundreds, even thousands, of other cells.

GRAY MATTER

The brain is divided into gray and white matter. Gray matter is made of neuron cell bodies, common in the surface of the brain. White matter is made of these neurons' myelinated axons bundled into tracts. They run through the middle of the brain and down the spinal cord.



Dendrites act like antennae to collect signals from neighboring nerve cells

Electrical pulse jumps from one myelin segment to the next, speeding up nerve signal

Axons can be several centimeters long

Dendrites are shorter than axons, usually up to only 16 hundred thousandths of a foot

Axon delivers signal from neighboring cell



**THE HUMAN
BRAIN CONTAINS
APPROXIMATELY
86 BILLION NEURONS**